

EE6320 RF Integrated Circuits
Project: VCO Design

VCO Performance Summary Table

Design Parameter	Design Metric	Performance	Specification
Output Amplitude (V _{pp} /2)	freq = 6900 MHz	2.489 V	≥ 1.2 V
	freq = 7000 MHz	2.539 V	≥ 1.2 V
	freq = 7100 MHz	2.62 V	≥ 1.2 V
Phase Noise (1 MHz Offset)	freq = 3450 MHz	-132.047 dBc/Hz	≤ -122 dBc/Hz
	freq = 3500 MHz	-132.668 dBc/Hz	≤ -122 dBc/Hz
	freq = 3550 MHz	-129.197 dBc/Hz	≤ -122 dBc/Hz
Phase Noise (20 MHz Offset)	freq = 3450 MHz	-155.378 dBc/Hz	≤ -154 dBc/Hz
	freq = 3500 MHz	-155.377 dBc/Hz	≤ -154 dBc/Hz
	freq = 3550 MHz	-154.139 dBc/Hz	≤ -154 dBc/Hz
Tuning Range	Total Tuning Range	215.34 MHz	≥ 200 MHz
	Number of bits in coarse tuning	8	-
	Voltage range in fine tuning	0.15 V - 1.05 V	-
	Average K _{VCO}	55.234 MHz/V	70 MHz/V
	% variation in K _{VCO}	12.72 %	minimal
Power Consumption (7000 MHz)	VCO average power consumption (excluding bias)	17.004 mW	minimal

	Bias circuit	22.8 mW	minimal
Other	Sum of all capacitances (in capacitor bank)	96 fF	-
	Net inductance used	2.6 nH	-
	Simulator used	Spectre	-

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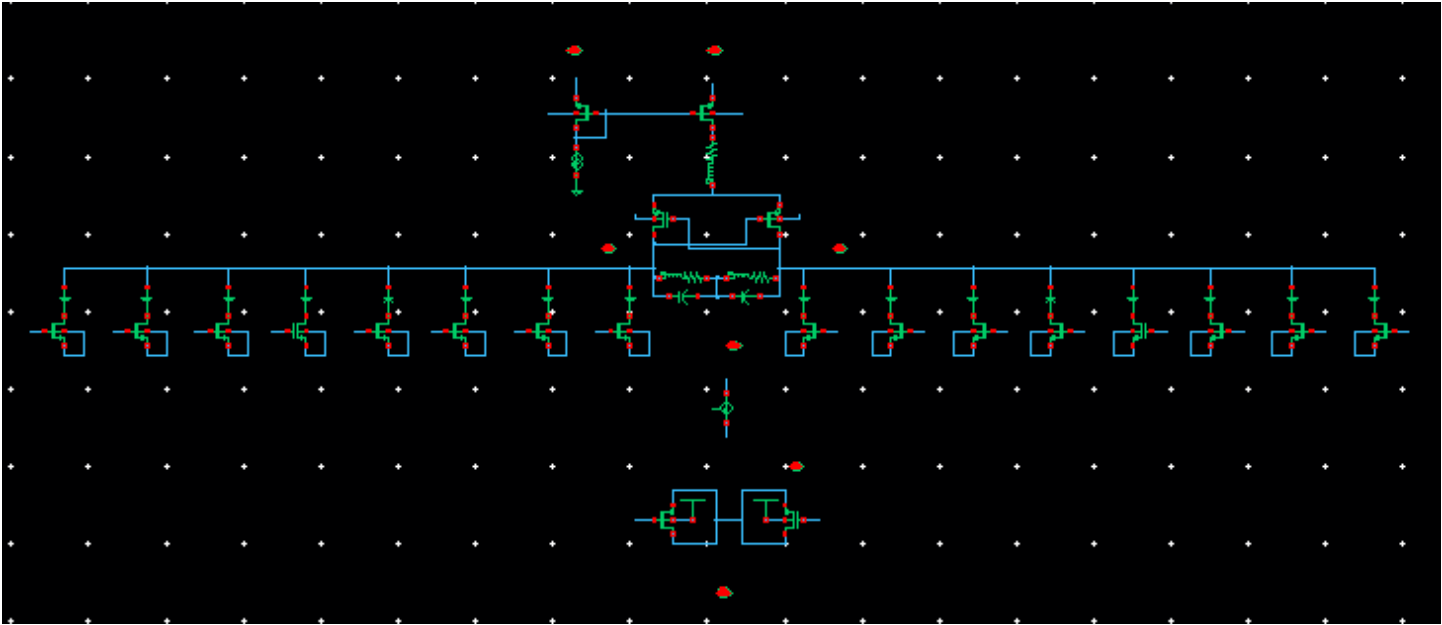
Component Values Table

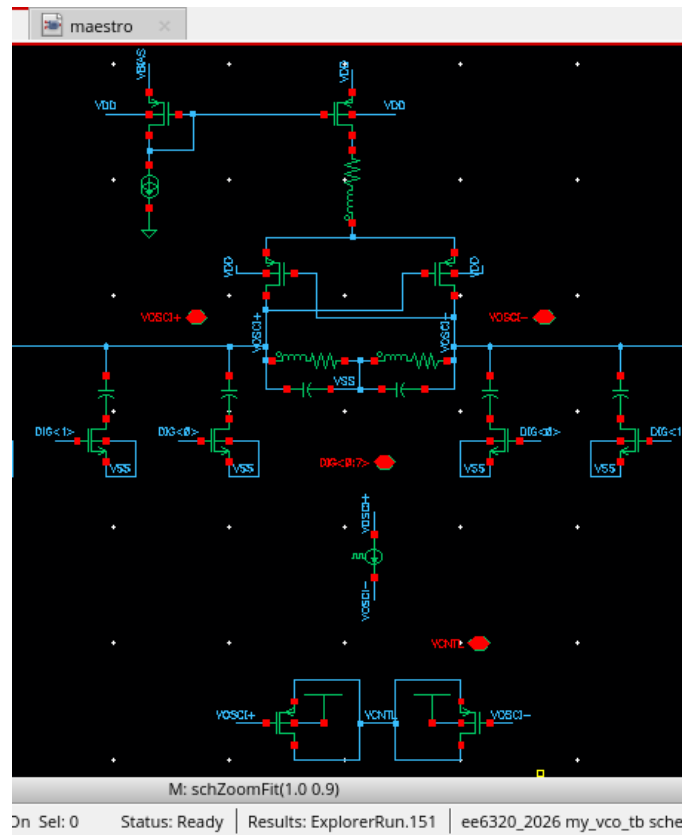
Component	Value <report only those which are used by you>
(W, L) Current Mirror MOS	(300*1 um, 0.12 um)
(W, L) Tail MOS	(300*1 um, 0.12 um)
(R, C) Low Pass Noise Cancelling Filter	-
Tail Capacitance	-
(L, C) Tail Tank	(2 nH, -)
(W, L) Cross Coupled MOS(s)	(120 um, 0.12um)
(W, L) Switching MOS(s)	(20*0.16 um, 0.12 um)
(C0, C1, C2, C3, C4, C5, C6, C7) Capacitor Bank	(12 fF, 12 fF, 12 fF, 12 fF, 12 fF, 12 fF, 12 fF, 12 fF)
(W, L) Varactor MOS	(50 um, 0.12 um)
(R, L) Drain Tank	(84Ω, 0.6 nH)
(V bias, I bias)	(1.2 V, 16 mA)

Divide by Two circuit Component values

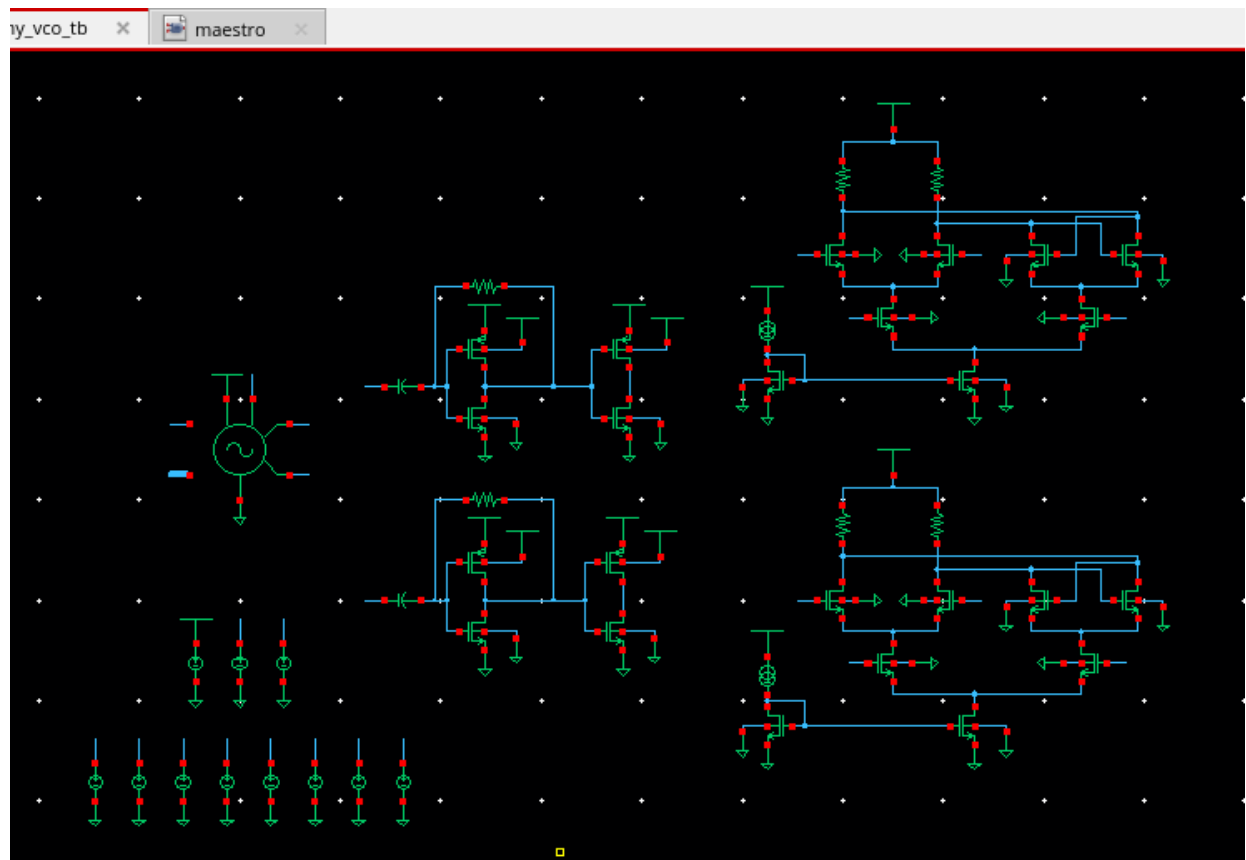
Component	Value
(W, L) Current Mirror MOS	(100*0.16 um, 0.12 um)
(W, L) Tail MOS	(100*0.16 um, 0.12 um)
(W, L) all MOSs	(4 um, 0.12um)
(R _L) Load Resistance	2.5 kΩ
(V bias, I bias)	(1.2 V, 1.5 mA)

Schematic



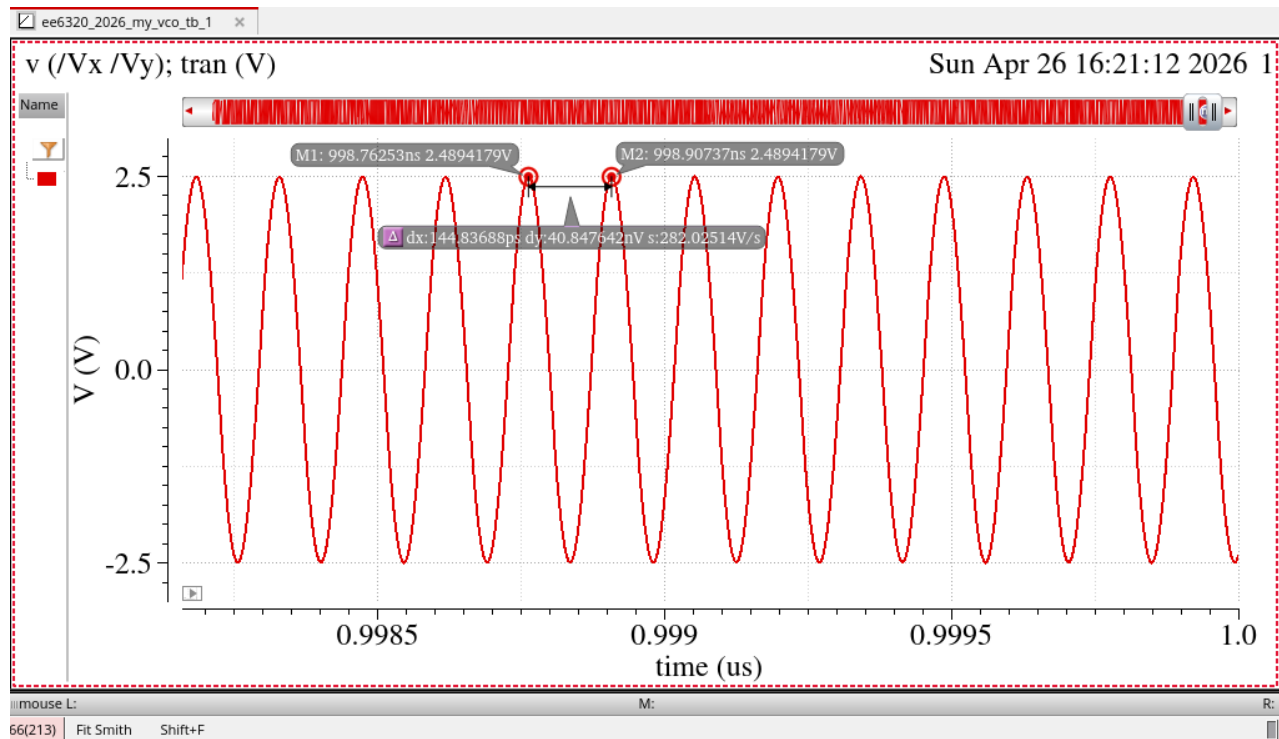


Testbench

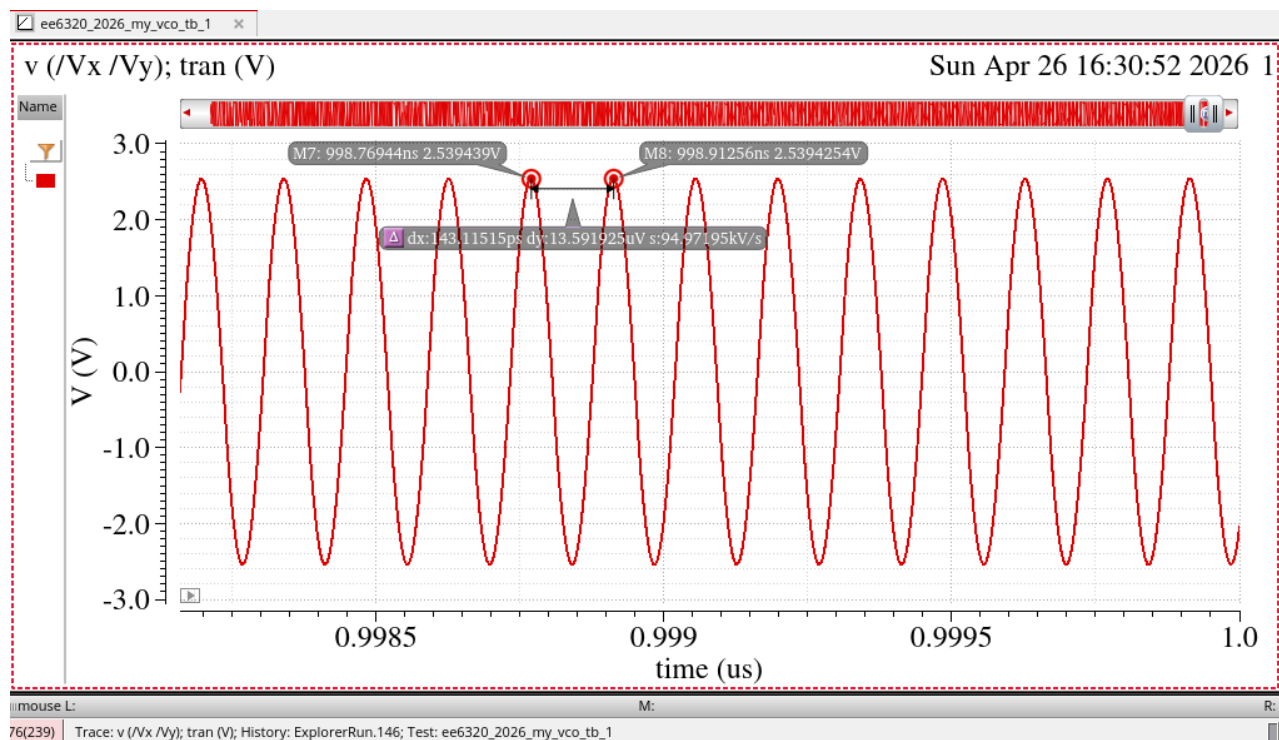


Output Amplitude Plots (differential settled waveforms)

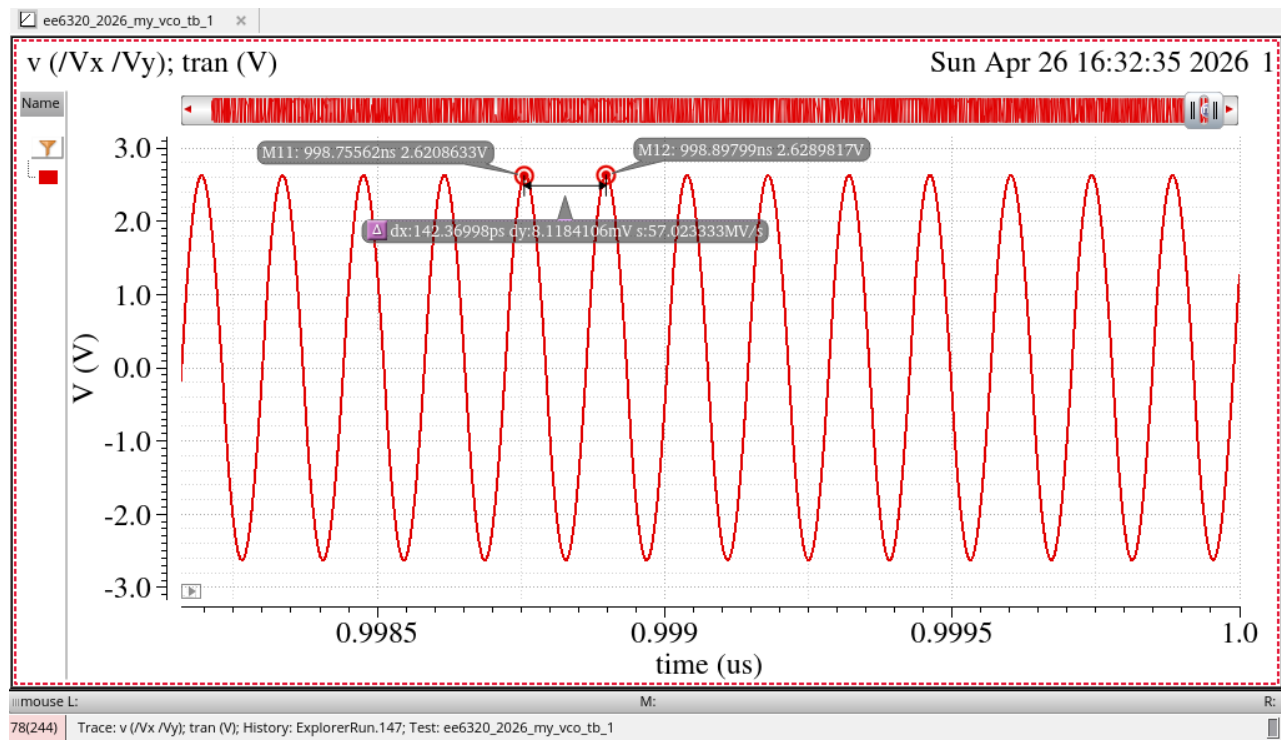
At 6900 MHz



At 7000 MHz

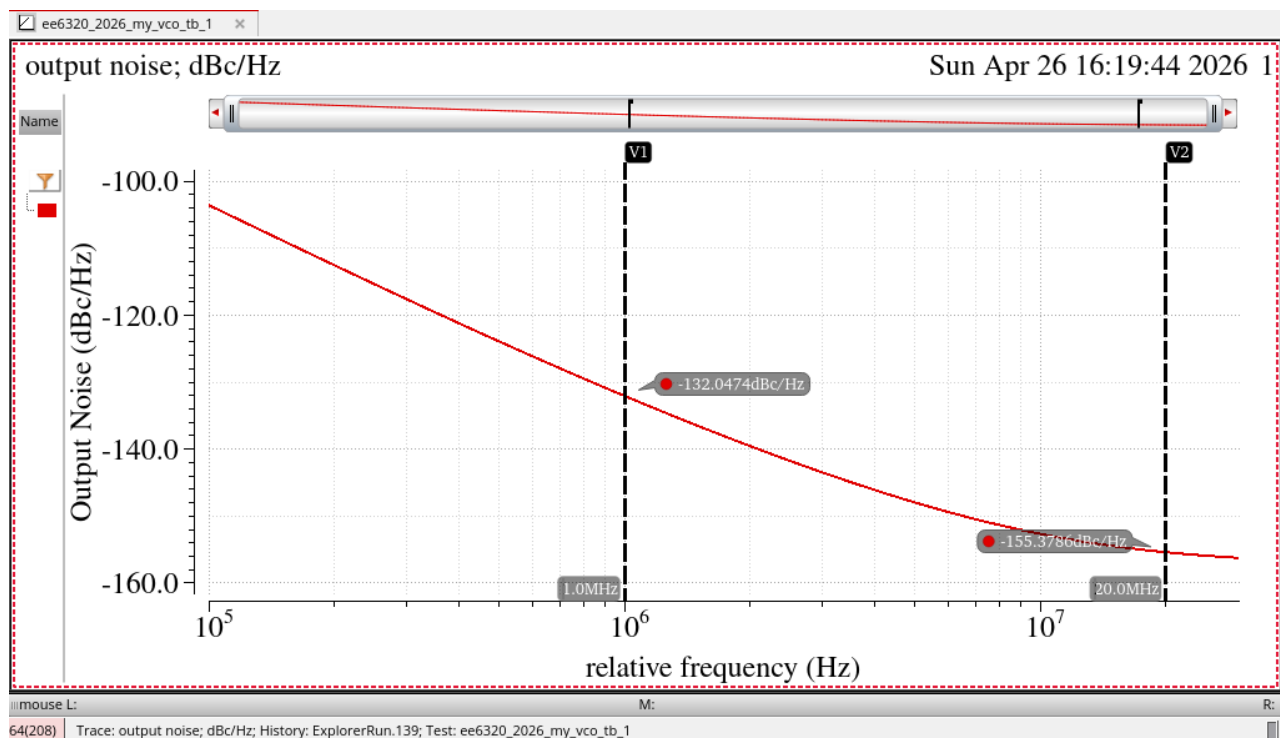


At 7100 MHz

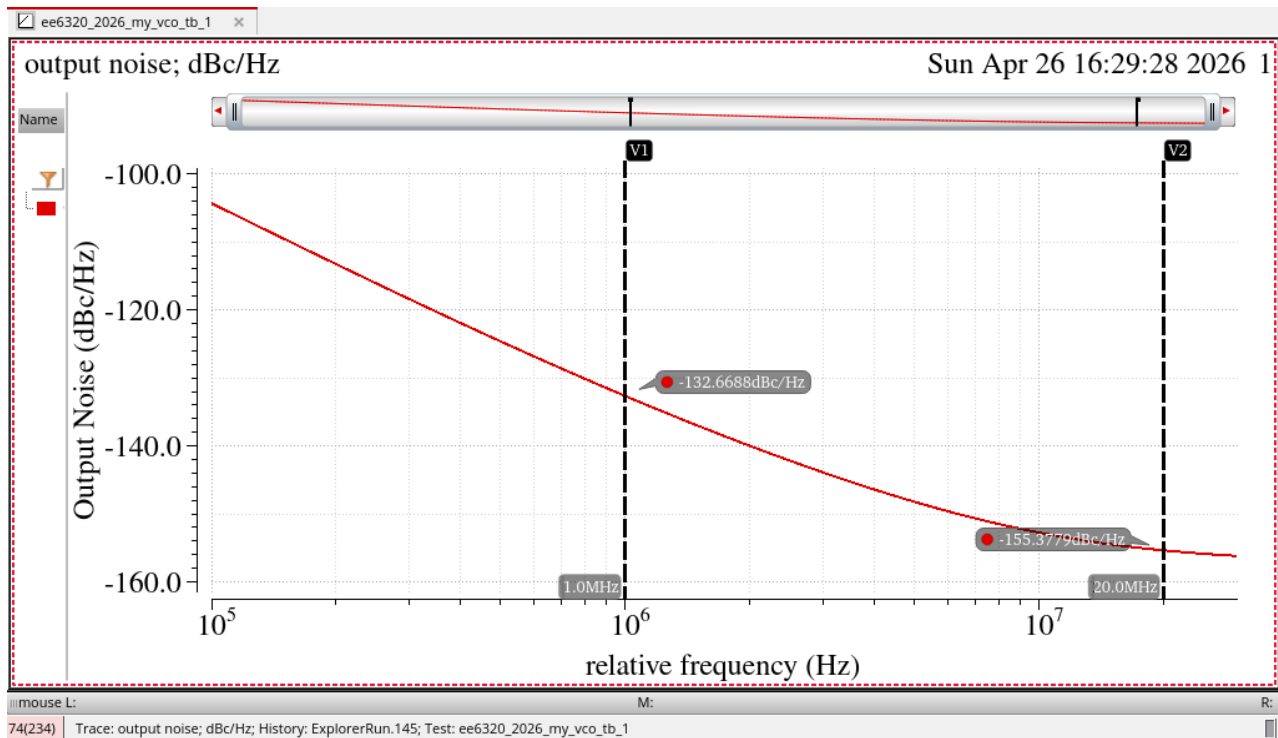


Phase Noise Plots (at 1 MHz and 20 MHz)

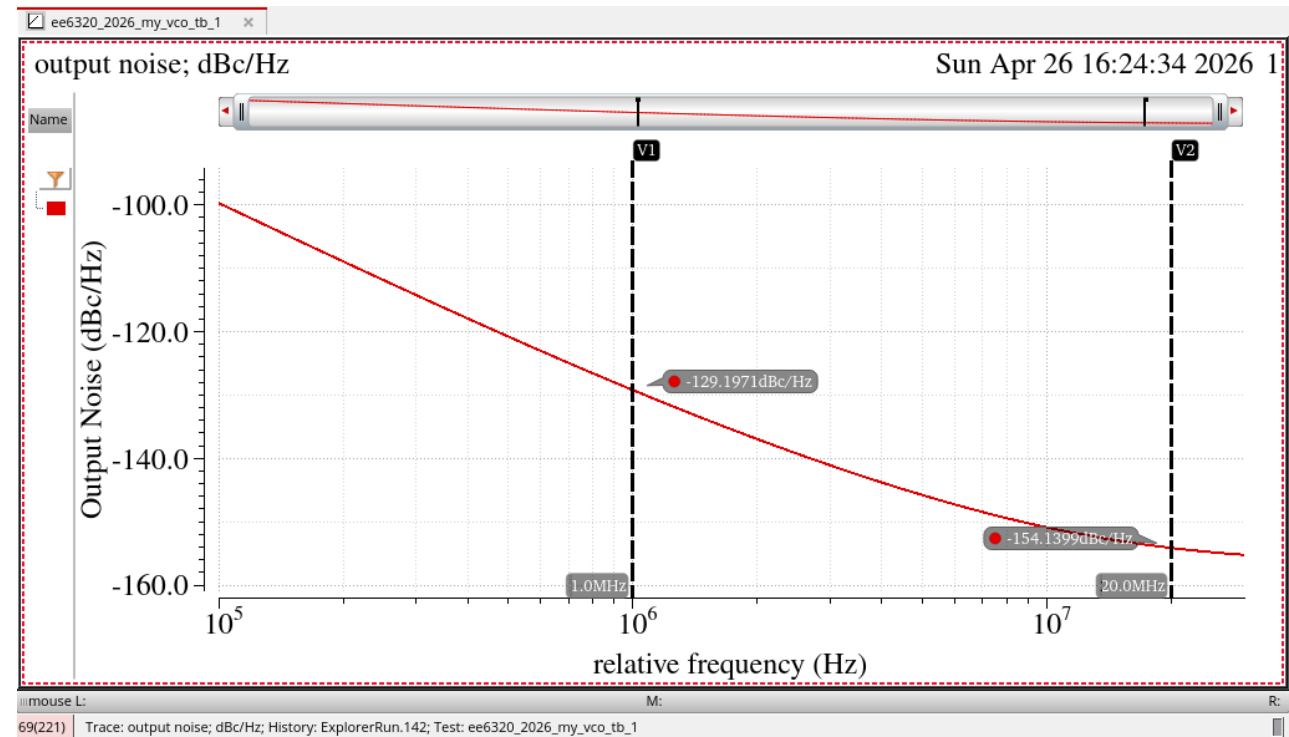
At 3450 MHz



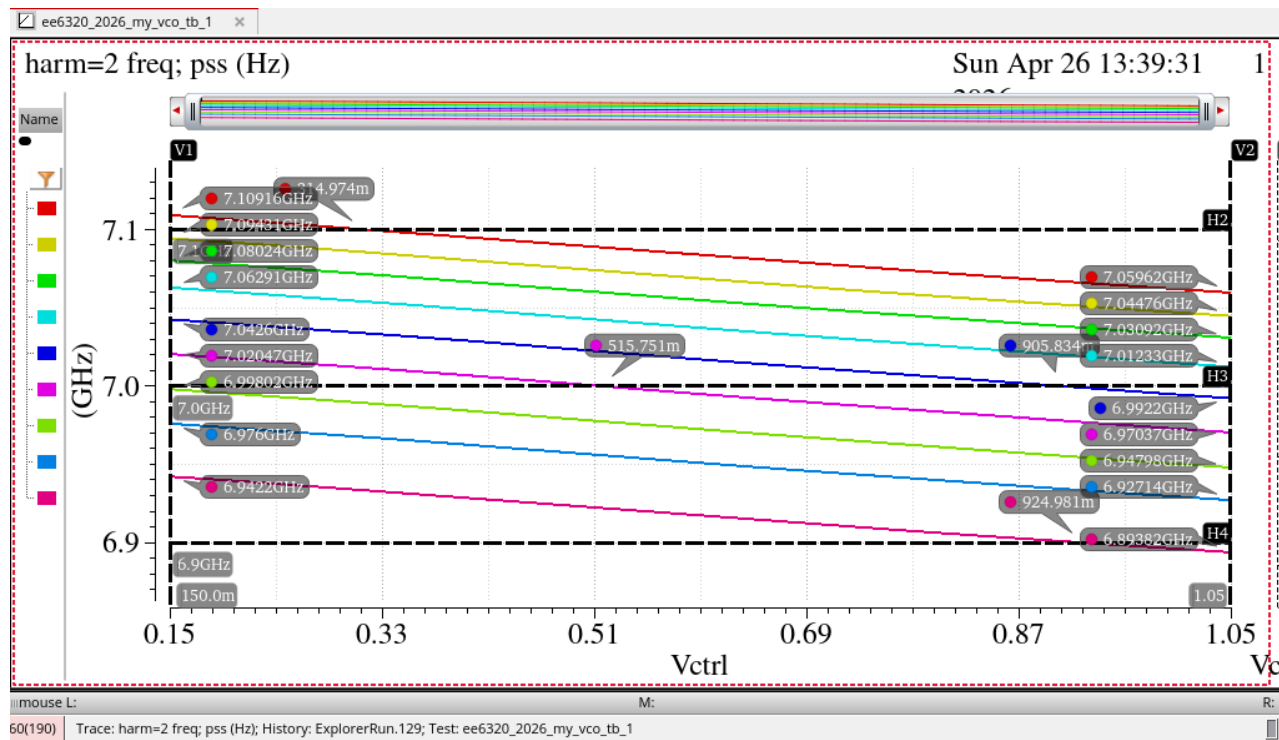
At 3500 MHz



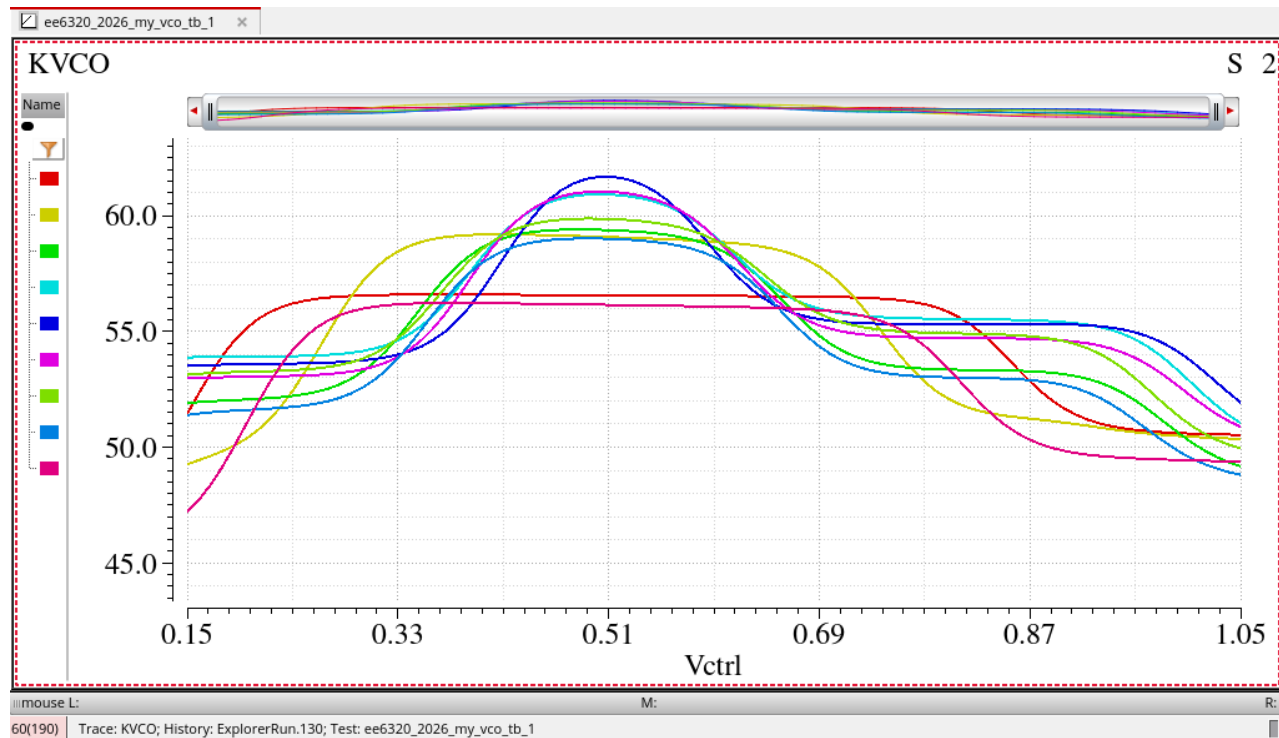
At 3550 MHz



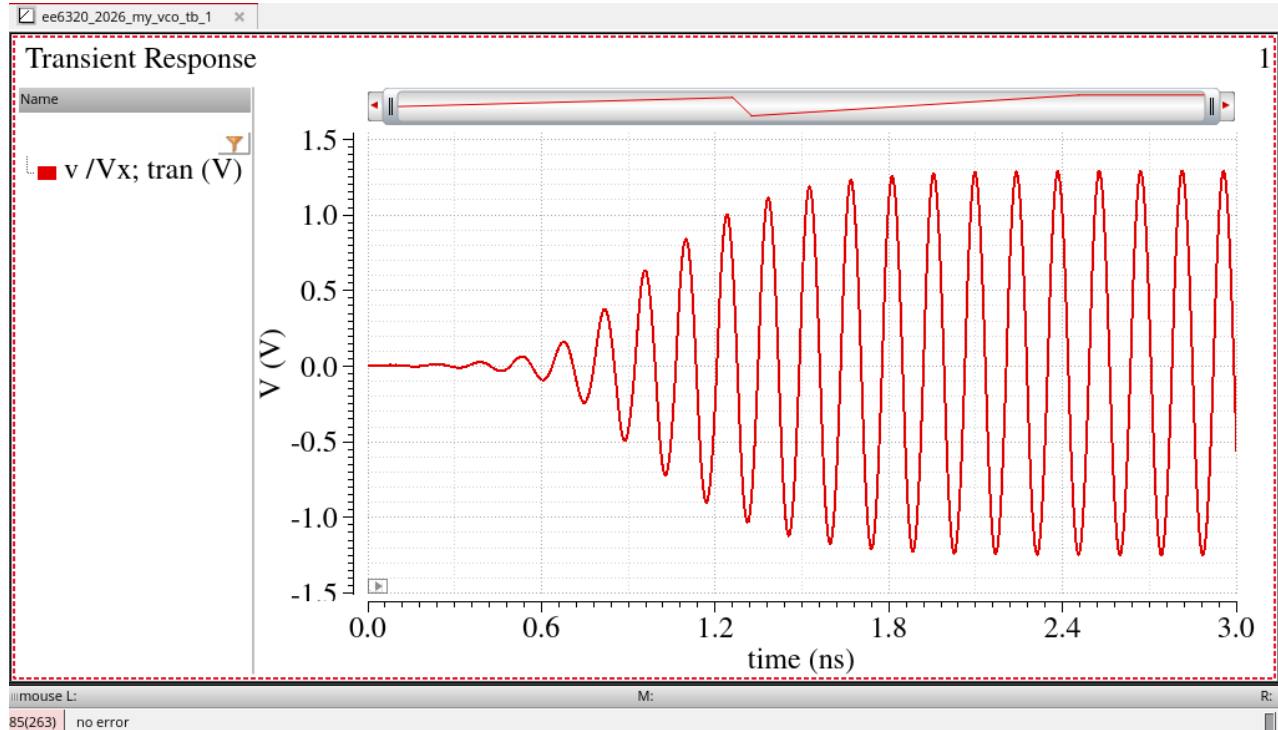
Frequency Tuning Range Plot



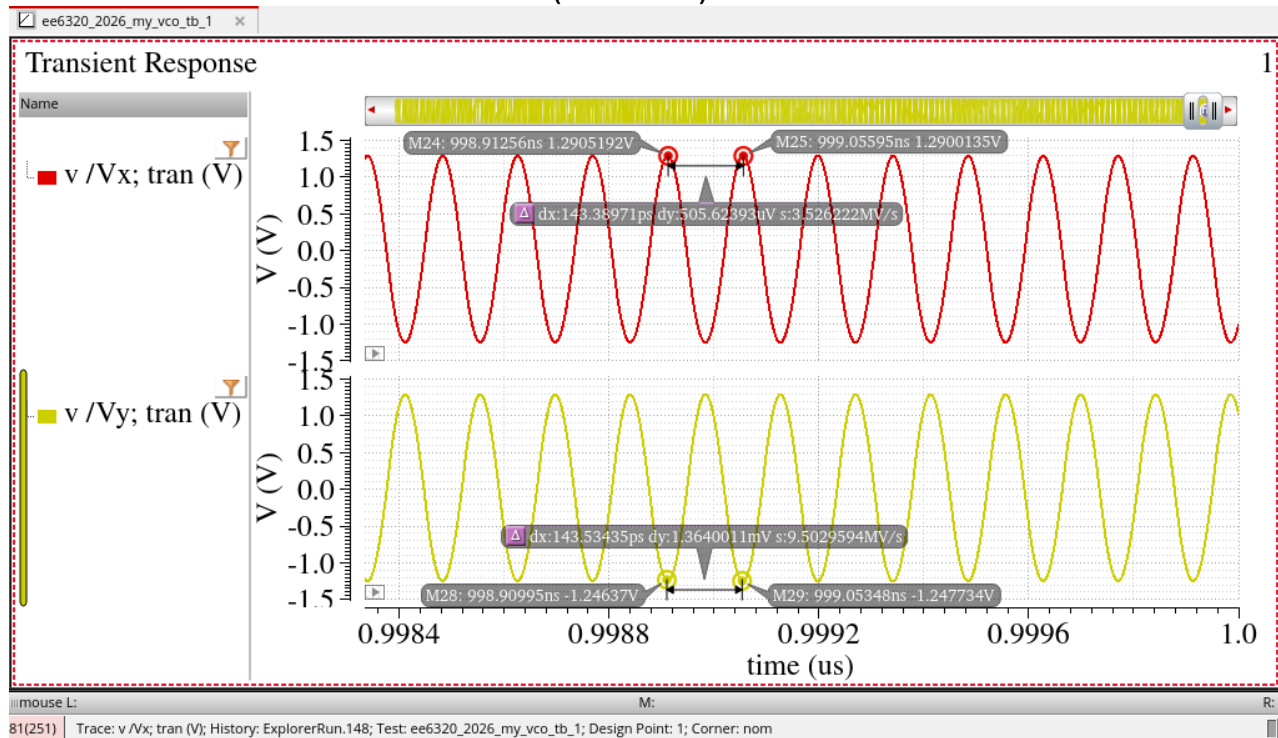
KVCO Plot



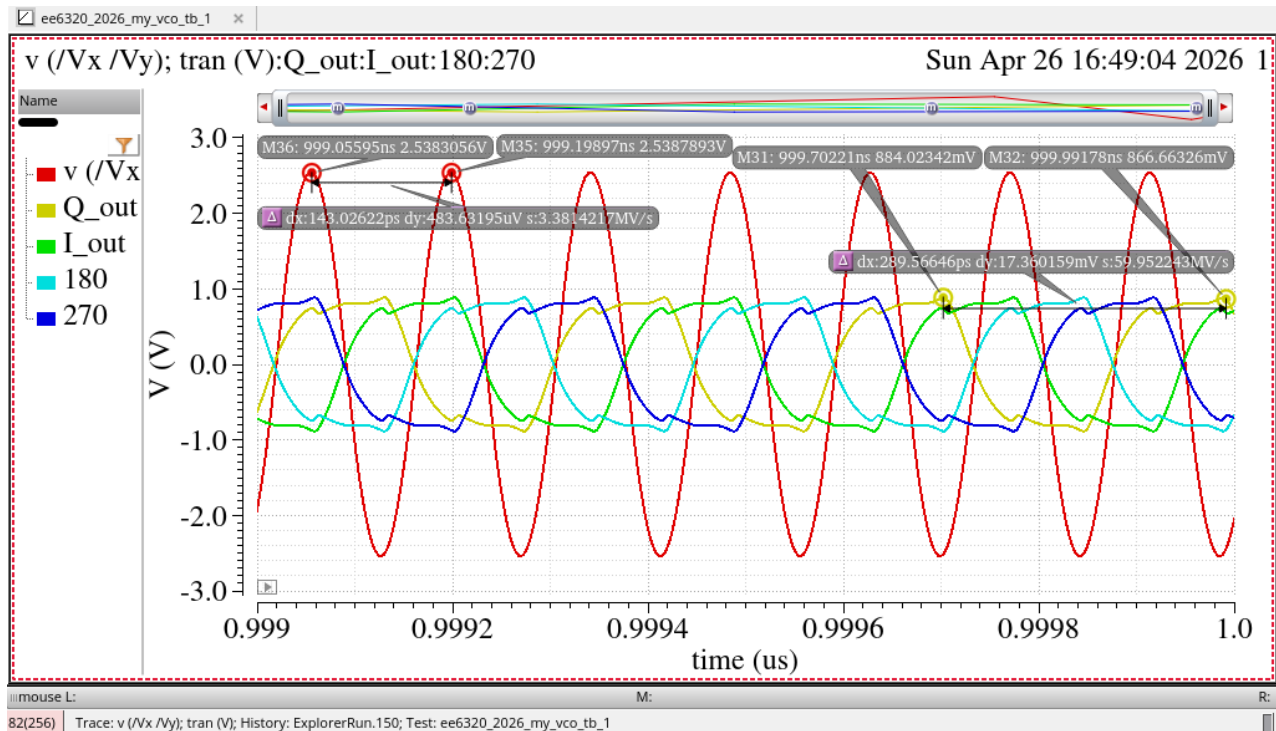
Initial Transient Response Plot (for Vop at 7000 MHz)



Single Ended Output Plot (Vop & Von) (at 7000 MHz)



Plots for 0, 90, 180 and 270 degree signals of Frequency Divider output



Design Procedure

- First, we set the values of the bias current and the drain inductor such that oscillations are seen ($g_m > 2/R$).
The MOSFETs should be sized such that no transistor is in triode.
Next, we set the drain capacitor to a value that causes the oscillations to be close to the required frequency, 7 GHz.
Then, we tweak the bias current or inductor value such that the amplitude of oscillation is higher than the minimum specified amplitude.
- Then, we connect a varactor and size it such that the KVCO is around 70 MHz/V.
- Then, we calculate the minimum number of capacitors we need in the bank, depending on the range of control voltage over which the KVCO is close to 70 MHz/V.
For example, if the KVCO is 70 MHz/V over a control voltage range of 600mV, then each frequency range would be 42 MHz wide.
To cover a tuning range of 200 MHz, we would have to divide it into 7 parts keeping in mind the constraint of the 33% overlap (0-42, 28-72, 56-98, 84-126, 112-154, 140-182, 168-210).
For this, we would need 6 capacitors if we had near perfect overlap.
- These capacitors, with values such that they take the frequency from 7.1 GHz when all are OFF, to 6.9 GHz when all are ON are then connected with MOSFETs acting as switches.
These MOSFETs need to be sized such that their parasitics do not affect the effective value of the capacitance that actually gets added to the oscillator.
- We also need a buffer to re-bias the output sinusoids, such that they can be fed into the frequency divider.
This can be an inverter that is self-biased.

It can also be followed by another inverter, in case the waves at the output of the first inverter are not fully square waves.

- Then, we make the frequency divider.

In this case, a current mirror latch, since it is inherently differential and is meant for such high frequencies.

We need to set the bias current and sizes of the MOSFETs such that the transistors are able to drive enough current in time, to bring down the output voltage and hold it at that value.

- Then, we plot the phase noise.

If it does not meet the specification, the noise summary page points out the main contributors to the output noise.

The following trends can be applied to reduce contributions due to certain components –

- The noise due to the transconductors can be reduced by moving to a higher operating current.
- An inductor can be added between the tail MOSFET and the transconductors to reduce its noise, with value such that the noise is minimum.
- The bias current can be increased to decrease the noise from the current mirror MOSFET.
- The sizes of the buffer's inverters can be increased to reduce their noise.
- The output resistance of the current mirror latch can be swept to find the point with the minimum possible noise.

Path to Project Files

`/home/ep22b020/cadence/EE6320_2026/ee6320_2026/ee6320_2026_vco`